FOR PROPER MAPS WITH CURVE FIBRES AND H^1 OF THE STRUCTURE SHEAF VANISHING, SURJECTIVITY OF STRUCTURE SHEAVES DESCENDS THROUGH ANY BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 21017-21027 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 21029-21056, the complete author proof (26 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The label's kappa suffix alludes to checking H^0 on residue fields; the statement is retained as written.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The curve-fibre vanishing package of the same chapter (card L16073) supplying H^1 vanishing for globally generated sheaves, flat base change for cohomology (Cohomology of Schemes, lemma-flat-base-change-cohomology), and the long exact sequence argument.

Source reference. Original lemma 02KH: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 21056–21056. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7H: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 20971–20980. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a proper morphism of schemes. Assume*

(1) *for all $y \in Y$ we have $\dim(X_y) \leq 1$ and $H^1(X_y, \mathcal{O}_{X_y}) = 0$, and*
(2) *$\mathcal{O}_Y \to f_*\mathcal{O}_X$ is surjective.*

Then $\mathcal{O}_{Y'} \to f'_*\mathcal{O}_{X'}$ is surjective for any base change $f' : X' \to Y'$ of f.

Proof. We may assume Y and Y' affine. Then we can choose a closed immersion $Y' \to Y''$ with $Y'' \to Y$ a flat morphism of affines. By flat base change (Cohomology of Schemes, Lemma 02KH) we see that the result holds for $X'' \to Y''$. Thus we may assume Y' is a closed subscheme of Y. Let $\mathcal{I} \subset \mathcal{O}_Y$ be the ideal cutting out Y'. Then there is a short exact sequence

$$0 \to \mathcal{I}\mathcal{O}_X \to \mathcal{O}_X \to \mathcal{O}_{X'} \to 0$$

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

where we view $\mathcal{O}_{X'}$ as a quasi-coherent module on X. By Lemma 0E7H we have $H^1(X, \mathcal{I}\mathcal{O}_X) = 0$. It follows that

$$H^0(Y, \mathcal{O}_Y) \to H^0(Y, f_*\mathcal{O}_X) = H^0(X, \mathcal{O}_X) \to H^0(X, \mathcal{O}_{X'})$$

is surjective as desired. The first arrow is surjective as Y is affine and since we assumed $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is surjective and the second by the long exact sequence of cohomology associated to the short exact sequence above and the vanishing just proved. $\square$